

Lokol Studio + Lokol Health

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PROBLEM

Because of Lokol Health, a nurse aide at a rural Solomon Islands clinic gets guidance from the national treatment manual, with the page cited, in Pijin or English, within a minute and without a signal. Today she would look it up late, skip it, or work from memory. 73% of Solomon Islanders are rural, nurse aides far outnumber doctors (524 vs 153), 3.5% of rural households have grid power, and fewer than half of people own a phone.

SOLUTION

Lokol Studio is a node-based builder for small offline AI helpers. Pick a sector, a language and the phone people actually carry; Studio recommends a graph of small models that fits the device's memory (speech in → guideline lookup → fine-tuned language model → rule-based safety gate → speech out), lets you swap or retrain any node, and exports a deploy pack: an offline PWA, an Android GGUF, a laptop/clinic-PC bundle, or a WhatsApp/SMS bot. **Lokol Health** is the first pack, grounded in the Ministry of Health's *Standard Treatment Manual for Children* (2017). It never diagnoses: it applies the manual, cites the section, refers danger signs (now, or on the next boat), and says "*Mi no sua, askem nes o dokta*" when the manual doesn't cover the question.

WHY AI, NOT SMS OR SEARCH

A static SMS tree or PDF search can't read a free-text or spoken Pijin description of a sick child, choose the manual section, spot a buried danger sign, and answer in Pijin. The parts that must be exact are not left to the model: a fixed danger-sign list forces referral, citations must be real sections, and anything off-script becomes "ask a person".

DATA AND TRAINING

2,704 synthetic training cases (plus 300 held-out test cases) in Pijin and English, generated from the manual by open-weight teachers (DeepSeek-V4.1-Flash, Kimi-K2.6) on River and validated against the manual. LoRA fine-tunes: Qwen3-0.6B and 1.7B with mlx-lm on a MacBook, Qwen3.5-9B on River; exported to 4-bit GGUF for llama.cpp.

STACK

Vite/React/React Flow app; in-browser runtime with llama.cpp in WebAssembly (wllama), transformers.js speech (MMS Pijin TTS as ONNX), BM25 retrieval over 56 manual sections, and the safety gate. Laptop tier adds llama-server, Omnilingual ASR for Pijin voice, and a FastAPI bridge for Twilio WhatsApp/SMS and Messenger. Hosted on Vercel; models on Hugging Face.

GUARDRAILS

- 12 danger signs force referral regardless of the model.
- No guideline match → "ask a person".
- No dose shown unless the cited page gives it for that drug.
- Data stays on the device; no diagnosis.

EVIDENCE (300 HELD-OUT CASES)

Model	Follows protocol	Right action (model / with gate)	Danger signs referred (gate)	Unsupported doses (model / gate)
Base models (0.6B, 1.7B, 9B)	0%	0% / 45%	75%	25-52% / 0%
Lokol Health 0.6B (397 MB, phone)	98%	64% / 62%	93%	88% / 0%
Lokol Health 1.7B (1.1 GB, phone)	99%	64% / 62%	95%	82% / 0%
Lokol Health 9B (River LoRA, laptop)	97%	87% / 84%	88%	63% / 0%

LIMITATIONS AND NEXT STEPS

Synthetic data not yet reviewed by a Pijin-speaking clinician; children's manual only; small models over-refer. Next: native-speaker and clinician review, field test with nurse aides, more manuals and sectors through Studio packs.

Live: lokol-studio.vercel.app · **Code:** github.com/VictorChenCA/lokol · **Models:** huggingface.co/VictorChenCA · **Dataset:** [data/synth](#) and [data/DATA_CARD.md](#) in the repo. Not a medical device; prototype only.